GOKULAHEALTH

User Research

Product Management Case Study | V1 → V2

Product Discovery & Research Artifact

Product	GokulaHealth
Artifact	User Research
Phase	Product Discovery
Version	V1 → V2
Research Status	Initial Discovery
Document Owner	Bhagyashree Reddy

1. Research Objective

The primary objective of this initial discovery research was to explore how small-scale cattle and dairy operators currently maintain milk-production records, understand potential operational difficulties associated with manually summarizing production performance, and gather preliminary feedback on whether automated calculations and summaries would provide value.

Specifically, as an early qualitative discovery study with a sample size of 1 prospective user, this study aimed to explore initial signals regarding:

- Daily logging preferences across morning and evening milking sessions.
- Time and effort involved in manually calculating weekly and monthly totals.
- Initial preferences for key production metrics (e.g., total volume, daily average, shift variance).

In addition, the research included an observation of the existing GokulaHealth V1 application to identify potential readability and clarity issues in its analytics experience. The findings are intended to inform initial hypotheses for GokulaHealth V2.

2. Research Context

GokulaHealth is an Android application designed for rural cattle management, livestock health tracking, milk-yield monitoring, and vaccination scheduling. In rural agricultural settings, intermittent internet connectivity makes offline capabilities relevant for digital tools.

Manual paper notebook record-keeping is commonly used in small-scale dairy operations, but it presents several potential limitations:

- **Physical Fragility:** Paper logbooks may be damaged, lost, or worn out in farm environments.
- **Calculation Overhead:** Summing daily yields over longer periods requires manual arithmetic.
- **Lack of Visual Insights:** Trends and shift comparisons can be difficult to observe from raw handwritten entries.

This research was conducted as an exploratory qualitative discovery study to gather initial feedback and mental models, rather than a statistical survey.

3. Target User

Target User Segments:

- **Primary Segment (Hypothesized Target):** Small-scale dairy operators and cattle owners responsible for daily operations, yield logging, health monitoring, and record management.
- **Secondary Segment (Farm Workers & Laborers):** Individuals who assist with daily milking and log morning and evening yields. Require simple and efficient data entry.
- **Secondary Segment (Family Members):** Assist with record checking, administration, or veterinary scheduling.
- **Potential Future Segment (Commercial Farm Managers):** Oversee larger herds requiring multi-tier reporting and batch analytics.

Product discovery focuses on addressing core usability and production logging needs for small-scale operators before exploring features for broader user segments.

4. Participant Profile

Participant Profile Details:

- **Status:** One prospective commercial dairy farm operator currently planning a cattle/milk business.
- **Context:** Planning operational workflows, record-keeping methods, and expectations for a future farm setup. Has not yet started active operations.
- **Relevance:** Provides initial prospective user expectations regarding record-keeping workflows and desired features.

Because the participant has not yet begun active farm operations, the findings reflect prospective expectations rather than observed real-world usage. These initial signals require further validation with active operators.

5. Research Method

Method: Early qualitative discovery discussion.

Sample size: 1 prospective user.

Research stage: Initial discovery.

The discussion focused on record-keeping habits, monthly production calculation, desired automation, and useful production information.

6. Areas Explored

How the participant expected to maintain cattle/milk-production records.

How monthly production calculations and analysis might be handled.

Whether automated calculations would provide value.

What production information would be useful to review.

Which production information should be considered highest priority.

7. Key Research Findings

7.1 Record-Keeping Approach: The participant indicated an intention to use physical paper notebooks for recording milk production. This provides an initial signal that manual/paper record keeping should be investigated further with active operators.

7.2 Monthly Calculation Effort: The participant expected manual monthly production calculations and multi-animal performance reviews to require time and effort.

7.3 Usefulness of Automated Summaries: The participant considered automated monthly summaries, averages, and production totals to be useful capabilities.

7.4 Priority Information: The participant expressed primary interest in monthly total production, daily/monthly averages, and morning versus evening production comparisons.

8. Pain Points & Observations Identified

Research Signals (From Participant Discussion):

- Manual paper-based record keeping may be cumbersome to maintain over time.
- Monthly calculation and summary analysis may require significant manual effort.
- Users may lack a quick way to review monthly milk performance.

Product Observations (From GokulaHealth V1 Review):

- Certain text and background combinations in V1 reduce visual readability.
- Analytics cards and chart components lack clear contrast and context.
- The 30-day production trend chart lacks axis context and clear metrics, making interpretation difficult.

9. User Needs

- An easy way to maintain daily milk-production records.
- A way to capture morning and evening production.
- Automatic calculation of useful monthly production summaries.
- A clear way to understand monthly total and average production.

- A clear comparison between morning and evening production.
- Useful cattle-level comparisons where relevant.

10. Evidence vs. Hypotheses

Type	Observation / Finding
Research Evidence	One prospective user expected to maintain records in a notebook.
Research Evidence	The participant expected monthly calculation/analysis to take time.
Research Evidence	The participant considered automatic calculation/analysis useful.
Product Observation	V1 production/analytics experience can be difficult to understand.
Hypothesis	Manual record keeping is a significant problem across the target market.
Hypothesis	Monthly total and average are the most valuable production insights.
Hypothesis	Users will regularly enter morning and evening production.
Hypothesis	Offline-first capability is important to the target user group.

11. Prioritized User Needs

1. Maintain production records easily.
2. Understand monthly total production.
3. Understand monthly average production.
4. Compare morning vs. evening production.

Understand cattle-level production differences.

This prioritization is preliminary and should be revisited after broader user research and usability testing.

12. Research Insights

Insight 1 — Entry Convenience is Essential: Paper logging is simple to initiate. A digital solution must provide an easy and fast entry experience to encourage regular usage.

Insight 2 — Automated Summaries Offer Apparent Value: Automated calculation of totals, averages, and shift breakdowns addresses a key friction of manual record-keeping.

Insight 3 — Clarity in Analytics is Critical: Displaying production data requires clear visual layout, proper contrast, and explicit labels so that users can easily interpret the information.

Insight 4 — Signals Require Further Validation: Because findings are based on a single prospective user and product observations, all insights represent initial hypotheses that require validation with active operators.

13. Implications for GokulaHealth V2

- Prioritize a simple milk-production recording workflow.
- Make monthly total and average production immediately understandable.
- Provide a clear morning-versus-evening view.
- Improve analytics readability and interpretation.
- Avoid prioritizing advanced analytics before core production needs are validated.

Use further research and testing to determine whether cattle-level and factory-level comparisons are valuable.

14. Open Research Questions

- How do active cattle/dairy operators currently record milk production?
- How often do they calculate monthly production totals and averages?
- How much time does monthly calculation currently take?
- Which production metrics do operators actually use to make decisions?
- Do users prefer daily, weekly, or monthly summaries?
- How many cattle are typically managed by the target user?
- Who enters the data: owner, family member, or farm worker?
- How important is offline operation in the user's actual environment?
- Would users consistently enter morning and evening production?

Would users find cattle-level comparisons useful for decision-making?

15. Research Gaps

- Only one participant has been consulted.
- No active cattle/dairy operator was observed using the current manual workflow.
- No usability testing of the V1 analytics experience has been conducted.
- No quantitative data exists yet on time saved through automation.

- No evidence yet establishes willingness to adopt or pay for the product.

Different user segments have not yet been compared.

16. Research-to-Product Decision

The research does not justify building every possible analytics feature. Instead, it supports a focused V2 hypothesis:

> ***IF GokulaHealth makes daily milk-production recording simple and provides clear monthly production summaries, THEN users may be able to review production performance with less manual effort and greater clarity.***

This hypothesis should be tested through additional discovery and usability validation before treating it as a confirmed product-market need.

17. Research Summary

Initial discovery suggests that manual milk-production record keeping and monthly calculation may create meaningful effort for cattle/dairy operators. One prospective user indicated that notebook-based records and monthly analysis could be time-consuming and that automated calculation would be useful.

The research points toward monthly total production, monthly average production, and morning-versus-evening comparison as promising areas for V2.

However, the evidence is currently limited to one prospective user. The next research stage should focus on validating the workflow, pain points, user segments, and value of the proposed production insights with additional real-world users.

17.1 Next Steps

Conduct additional discovery and usability research with active cattle/dairy operators to validate current assumptions before finalizing V2 requirements.

18. Document Control

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